

Automated Review Rating System

1. Project Overview

This project aims to automate the process of generating ratings from textual reviews. It involves developing an AI-based system, specifically utilizing deep learning, that automatically predicts the numerical rating (e.g., 1 to 5 stars) of a product, service, or business based solely on user-generated textual reviews.

Problem Statement: Manual review rating processes are prone to inconsistencies and manipulation, making it challenging for platforms to accurately gauge sentiment at scale.

Primary Goal: To accurately predict star ratings (1-5) from textual reviews.

Secondary Goals: To reduce manual effort, enhance feedback consistency, and provide scalable sentiment analysis for platforms.

Expected Outcomes: A robust system that provides reliable, automated rating predictions, fostering more trustworthy feedback ecosystems and aiding platforms in efficiently understanding customer sentiment.

2. Environment Setup

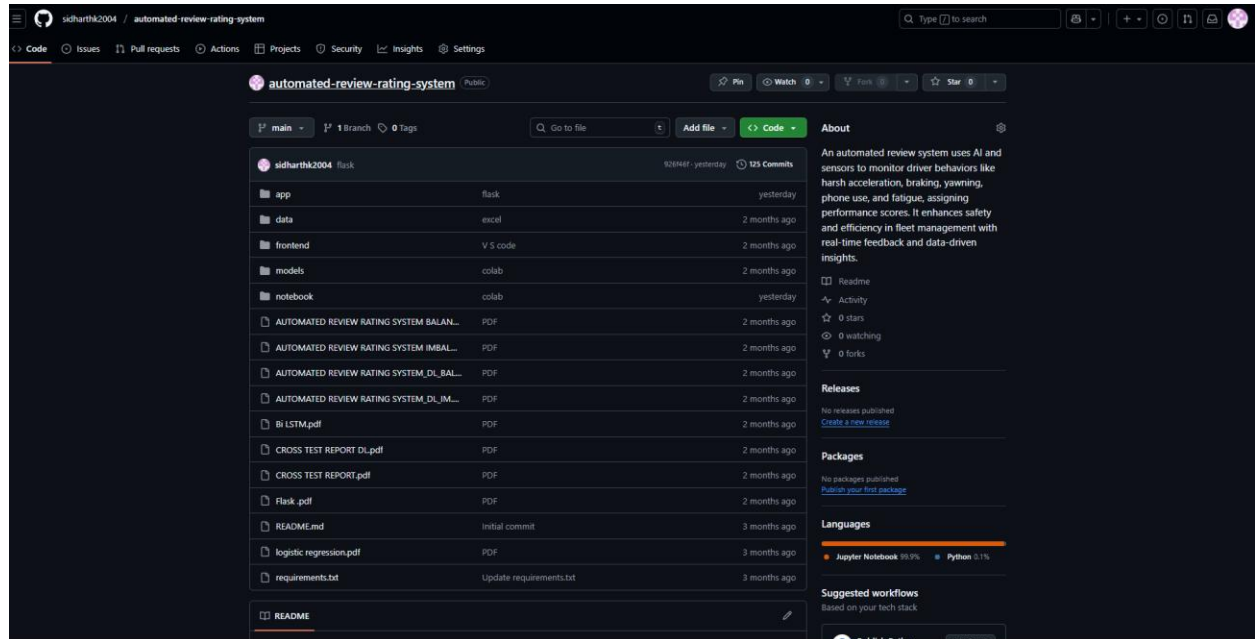
The development environment for this system was primarily Colab, providing access to necessary computational resources. The project was built using Python, with the following key libraries installed and utilised:

- **numpy:** Essential for numerical operations, especially with large datasets and array manipulations.
- **pandas:** Used extensively for data manipulation and analysis, particularly for handling DataFrames.
- **matplotlib** and **seaborn:** Employed for creating various data visualisations to understand data patterns and distributions.
- **scikit-learn:** A comprehensive library providing tools for machine learning tasks, including data splitting and potentially some preprocessing utilities.
- **nltk (Natural Language Toolkit)/Spacy:** Crucial for natural language processing tasks, such as text cleaning, tokenisation, and potentially stopword removal or lemmatisation.

(Note: Specific version numbers for Python and each library are recommended for full reproducibility but were not explicitly detailed in the provided sources.)

3. GitHub Project Repository

The project is hosted on a GitHub Project Repository. *(Note: A direct link to the repository, its high-level directory structure, and any contribution guidelines would enhance this section.)*



Github link: <https://github.com/sidharthk2004/automated-review-rating-system>

4. Data Collection

Data for the Automated Review Rating System was collected from Kaggle datasets. A total of 7 datasets were gathered, primarily consisting of reviews from Amazon in the Books domains. Each dataset contains two main columns: "review " and "rating" (1–5).

The selection of Amazon reviews from these domains provides a rich and diverse set of customer sentiments relevant to common product and media consumption.

Specific datasets collected include:

- **Amazon Books 1:** This dataset contains approximately 356588 reviews
- **Amazon Books 2:** This dataset contains approximately 1048574 reviews
- **Amazon Books 3:** This dataset contains approximately 474673 reviews
- **Amazon Books 4:** This dataset contains approximately 1282528 reviews

All 7 collected datasets were subsequently merged into a single dataset for comprehensive processing. This merging process involved concatenating the review and rating columns from each dataset to form a unified corpus.

5. Data Cleaning and Preprocessing

Preprocessing is a crucial initial step to transform raw data into a clean, structured format suitable for machine learning models. The following steps were performed:

- **Load and Inspect Datasets:**

- All 7 datasets were loaded, inspecting their structure, column names, data types, and sample records.
- Verification of "review" and "rating" columns was performed.
- Functions like `pd.read_csv()` and `pd.read_excel()` were used for loading.
- Commands such as `df1.head()`, `df1.tail()`, `df1.columns`, `df1.shape`, `df1.dtypes`, `df1.info()`, and `df1.describe()` provided initial insights into the data.

- **Remove Unnecessary Columns:**

- Columns not required for prediction (e.g. 'overall', 'vote', 'verified', 'reviewTime' etc.) were dropped to simplify the dataset and focus on relevant information.
- Only review and rating columns were retained.

- **Remove Null or Missing Values:**

- Rows containing null values in either the "review" or "rating" columns were removed. This step is crucial to prevent model errors and ensure data integrity, as missing values can lead to unexpected model behaviour.
- `df.isnull().sum()` was used to identify missing values, and `df.dropna(subset=['review_text', 'rating'], inplace=True)` was applied to remove them.

- **Remove Duplicates and Conflicting Reviews:**

- **Exact duplicates** (same review text and same rating) were handled by keeping only one instance to prevent biasing the model.
- **Conflicting reviews** (same review text but different ratings) were entirely removed. This decision implies that such contradictory entries are considered unreliable or ambiguous, and their inclusion could introduce noise or confusion for the model.
- `df.duplicated().sum()` checked for duplicates, and `df.drop_duplicates(inplace=True)` removed them.

- **Clean Text Data:**

- All text data was converted to lowercase to standardise text and reduce vocabulary size.
- The following elements were removed from the text: URLs, punctuation, numbers and special characters, and extra spaces. This ensures that the model focuses on the semantic content of the reviews rather than formatting or extraneous characters.

```
import re

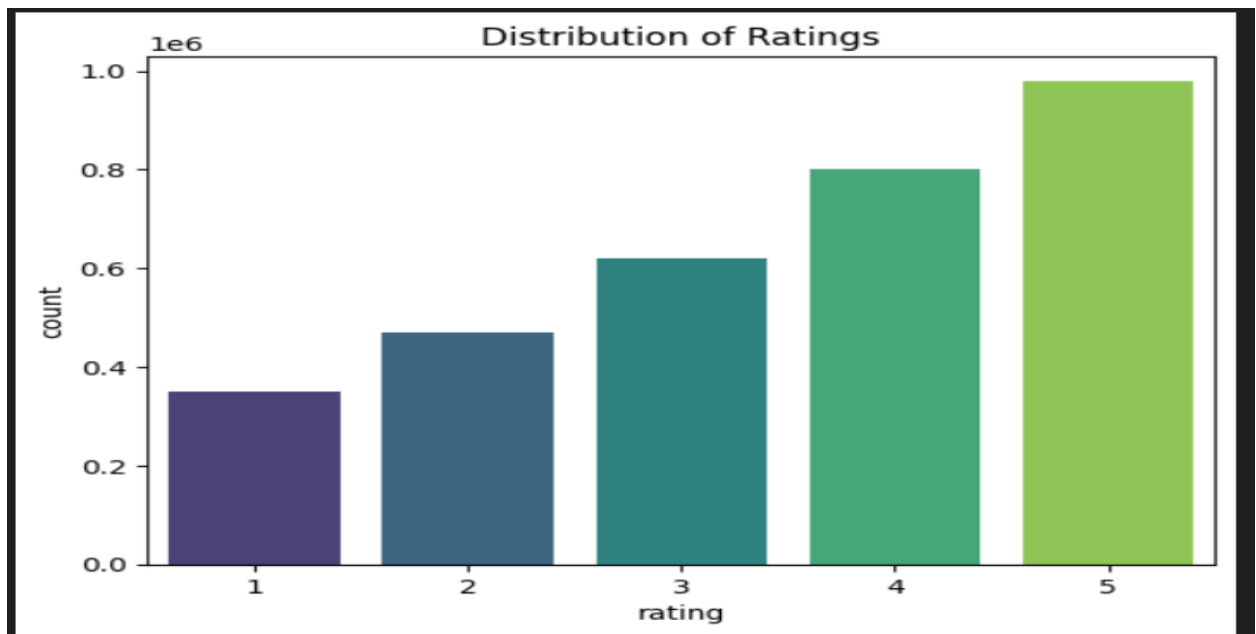
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"^\w\s", "", text)
    text = re.sub(r"\d+", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df.loc[:, "review"] = df["review"].apply(clean_text)
```

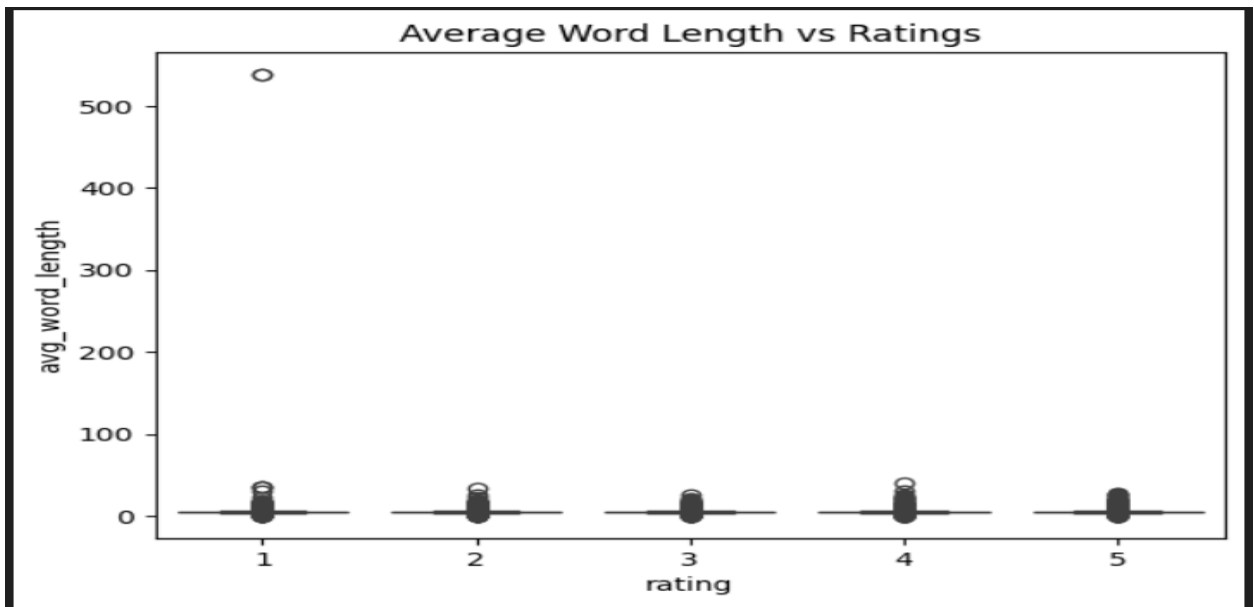
6. Data Visualization

Data visualisation was employed to understand patterns, trends, and insights within the data using visual elements.

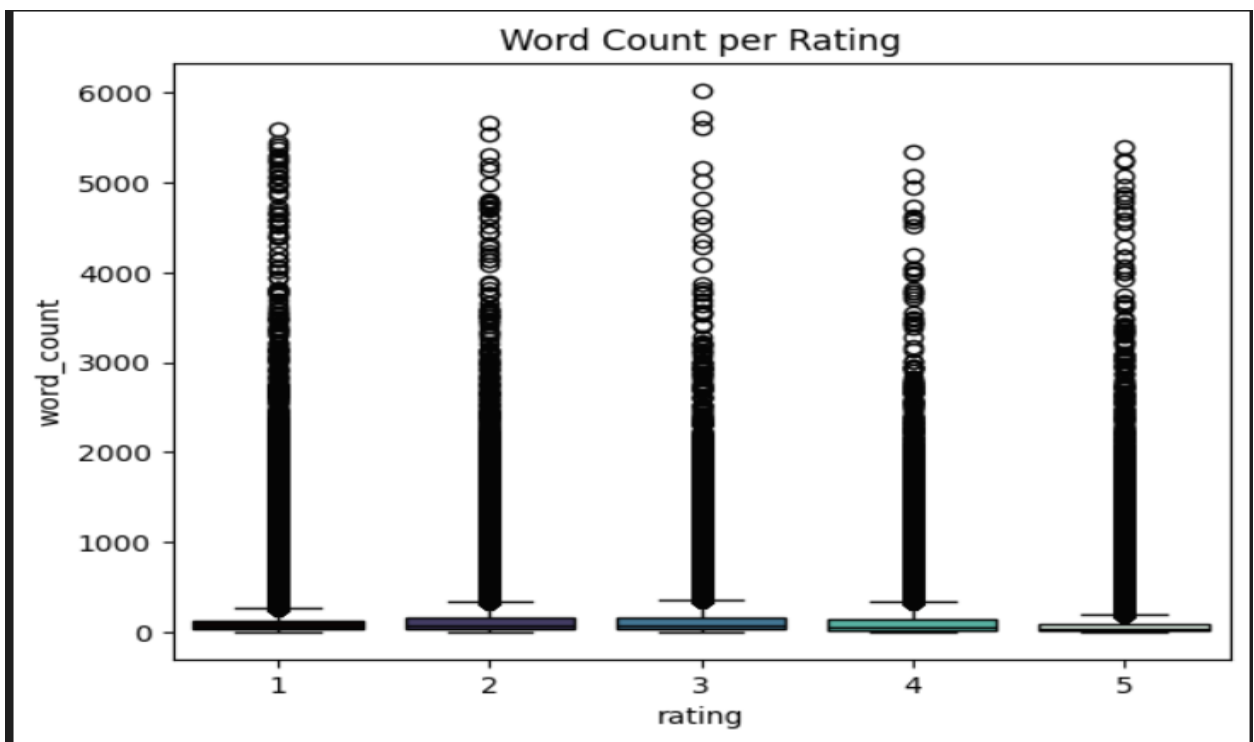
- **Bar Chart:** Used to show the number of reviews for each star rating (e.g., how many 1-star, 2-star, etc.). This helped to identify the distribution of ratings and potential class imbalances.



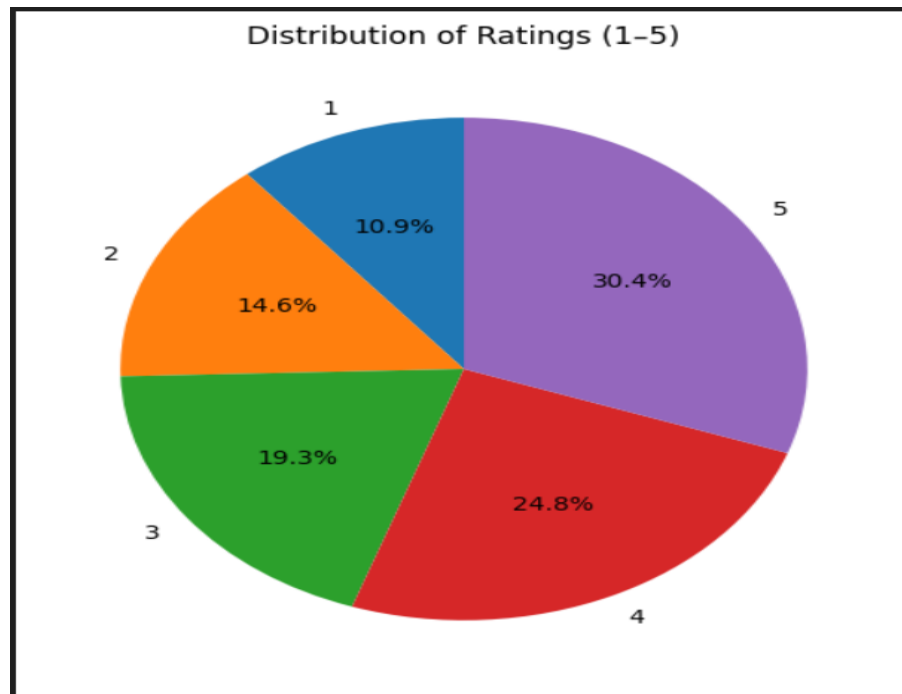
- **Box plot:** box plot that shows how the **average word length** (avg_word_length) varies across different **ratings** (rating) in the dataframe df, helping you visually compare the distribution and spread of word lengths for each rating category.



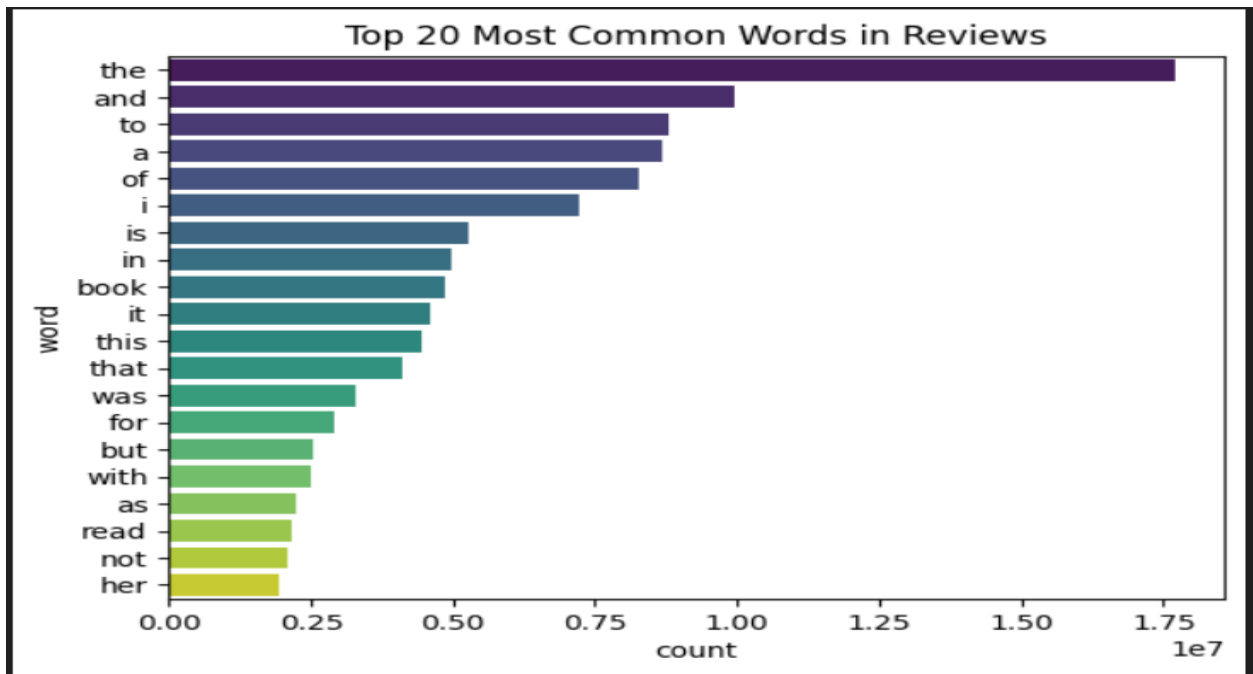
This code shows how long reviews (in word count) tend to be for each rating category, helping you see if higher or lower ratings usually come with longer or shorter reviews.



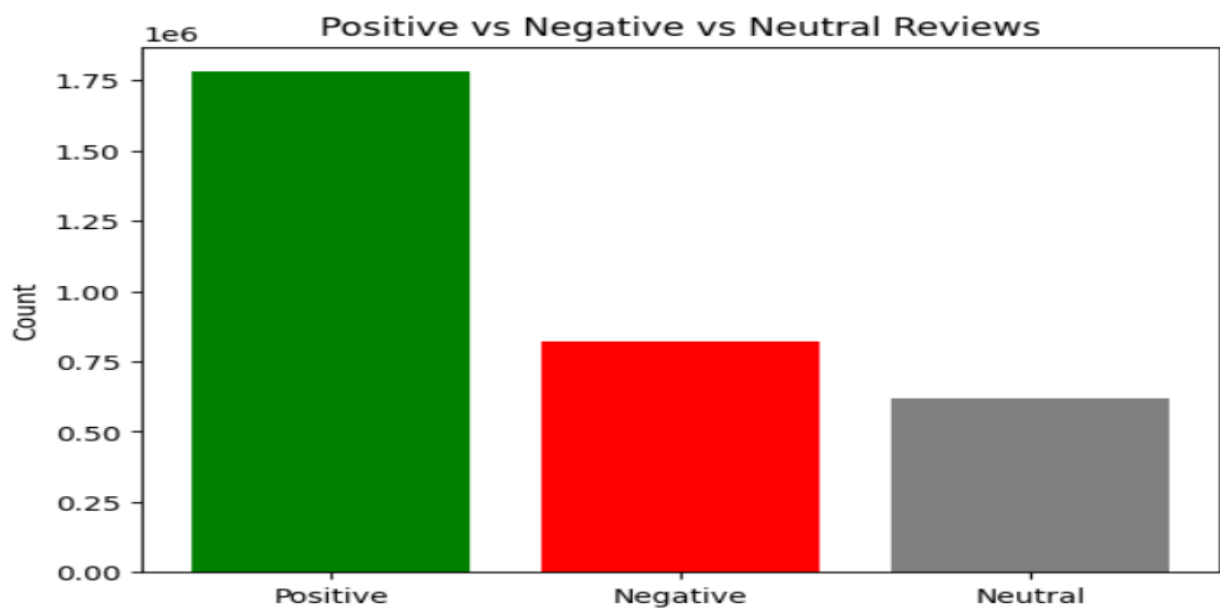
- **Pie Chart:** A circular chart divided into slices to illustrate proportions. Used to show the percentage distribution of each rating (e.g., how much of the data is 5-star, 4-star, etc.). This offered a clear overview of the overall rating proportions in the dataset.



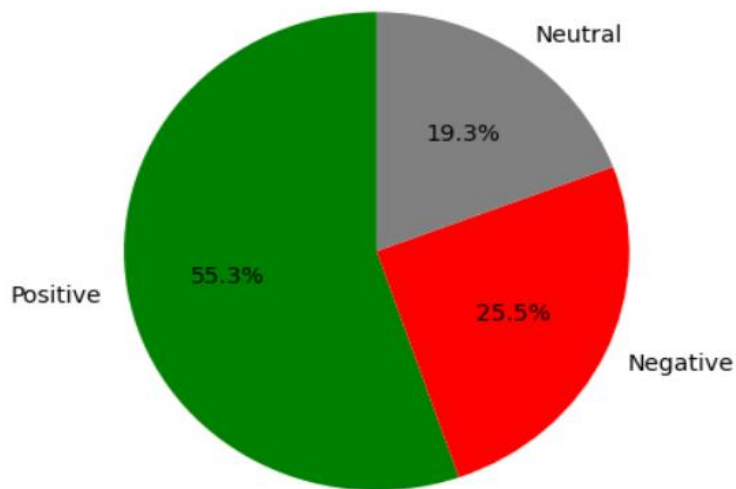
- **Most frequently used words:** [('the', 17694934), ('and', 9968206), ('to', 8787554), ('a', 8672555), ('of', 8278769), ('i', 7227150), ('is', 5272284), ('in', 4984232), ('book', 4853590), ('it', 4586243), ('this', 4444903), ('that', 4091459), ('was', 3281288), ('for', 2921000), ('but', 2519557), ('with', 2516753), ('as', 2232665), ('read', 2143519), ('not', 2081882), ('her', 1937802)]
- **Word frequency:** A word frequency visualization in review rating is a graphical representation that shows how often specific words appear in customer reviews, categorized by their rating scores (e.g., 1-star to 5-star). It helps uncover patterns in language usage across different sentiment levels.



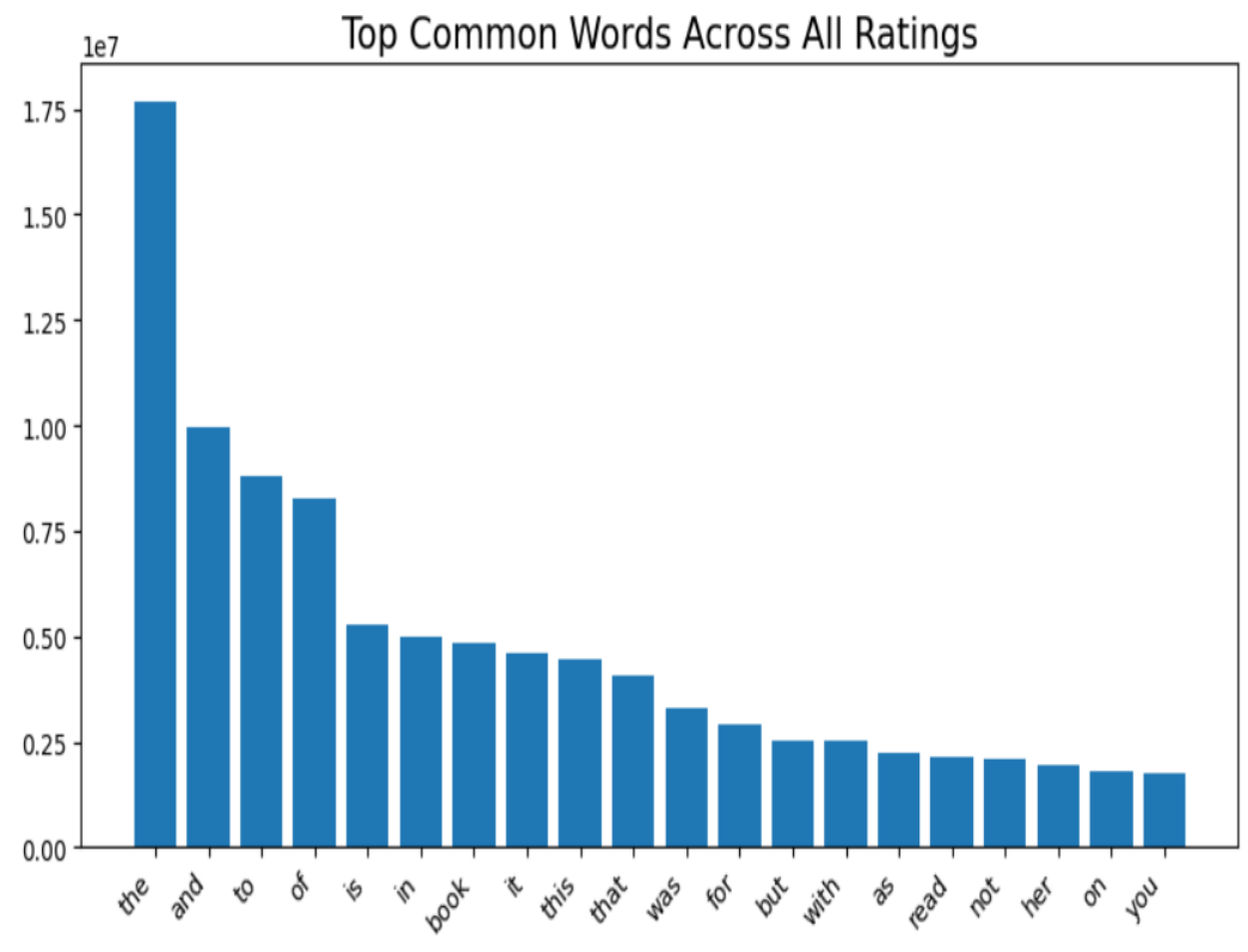
- Sentiment Distribution Visualization(Bar & Pie Charts):



Positive vs Negative vs Neutral Reviews

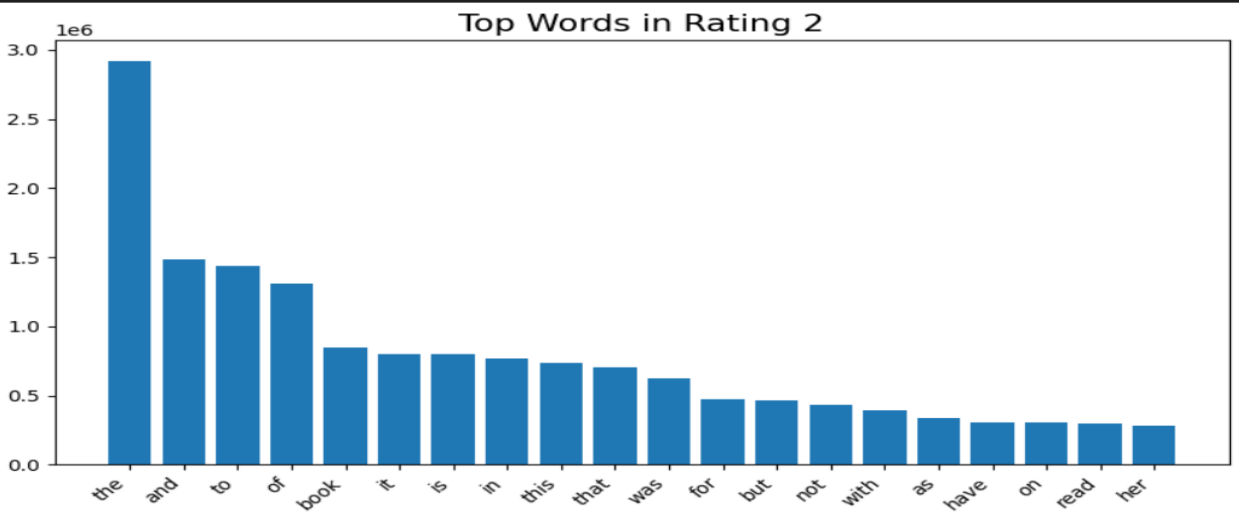
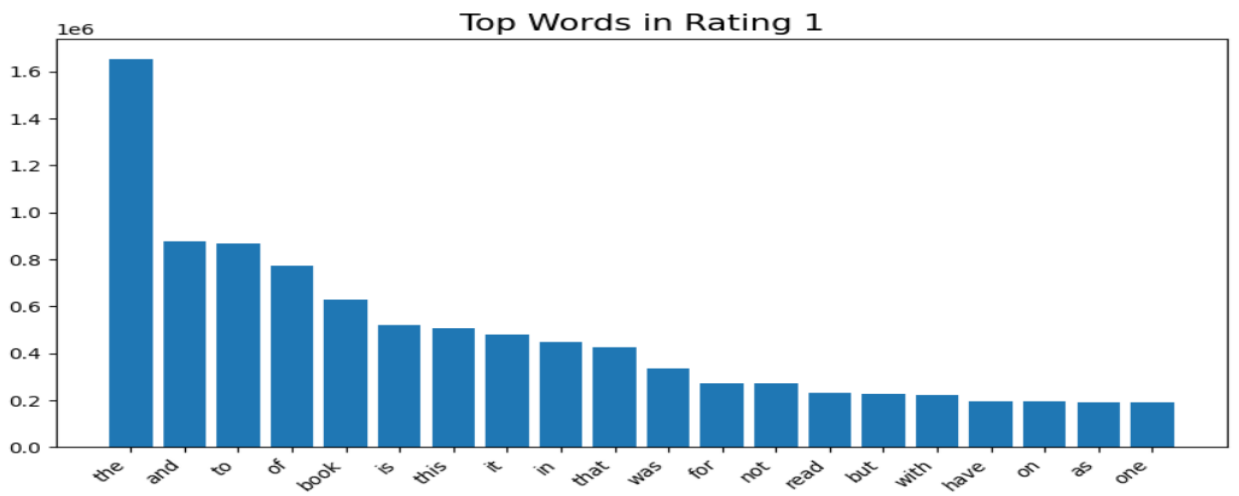
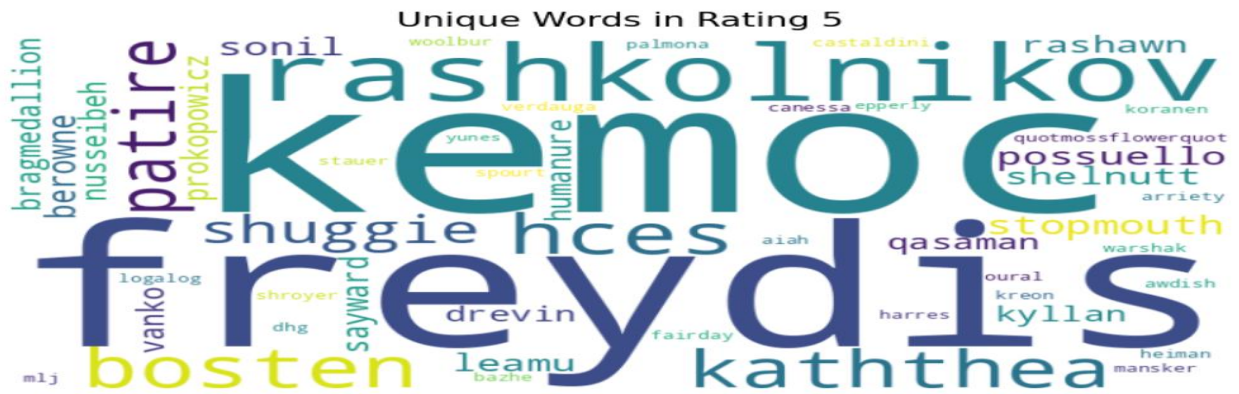


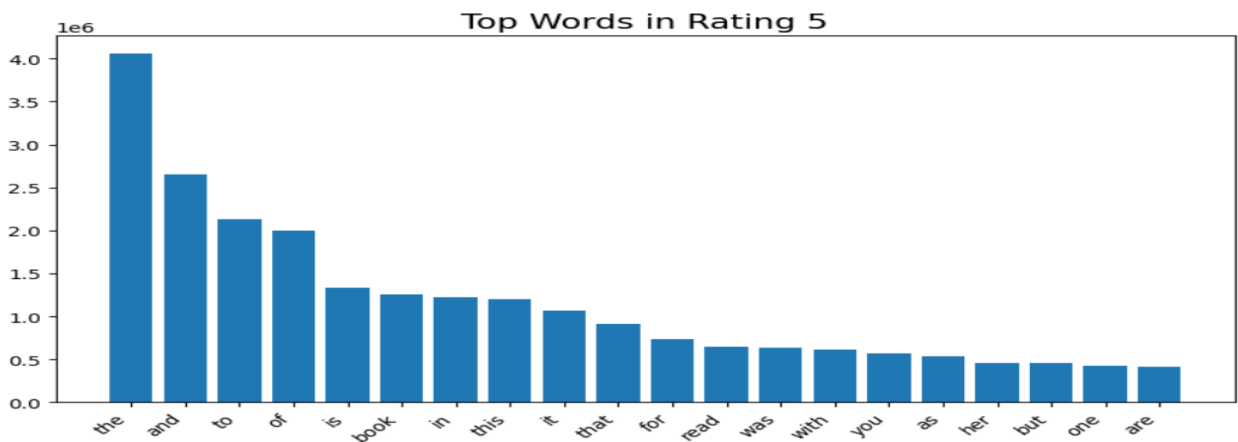
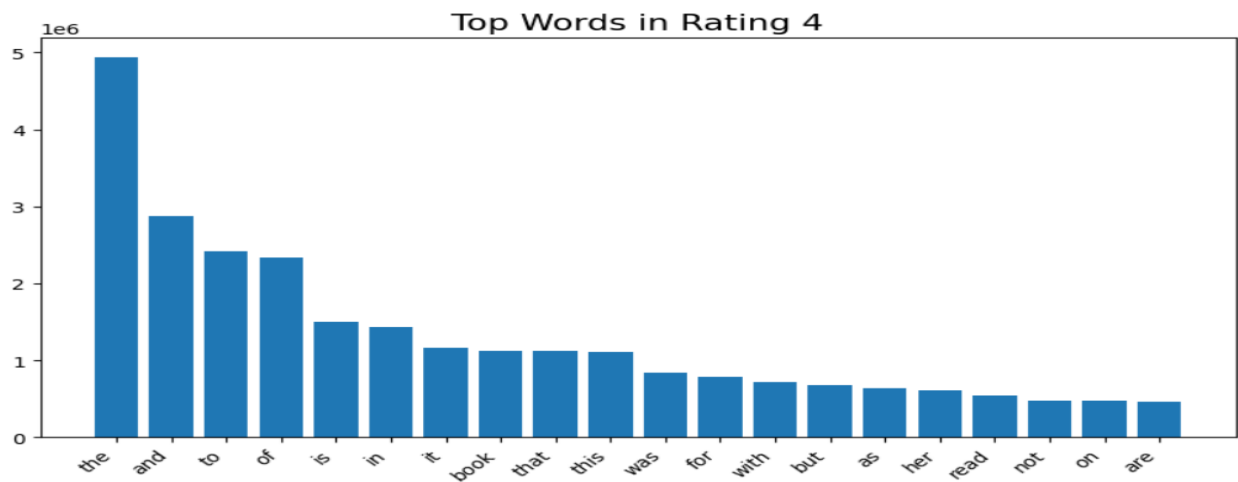
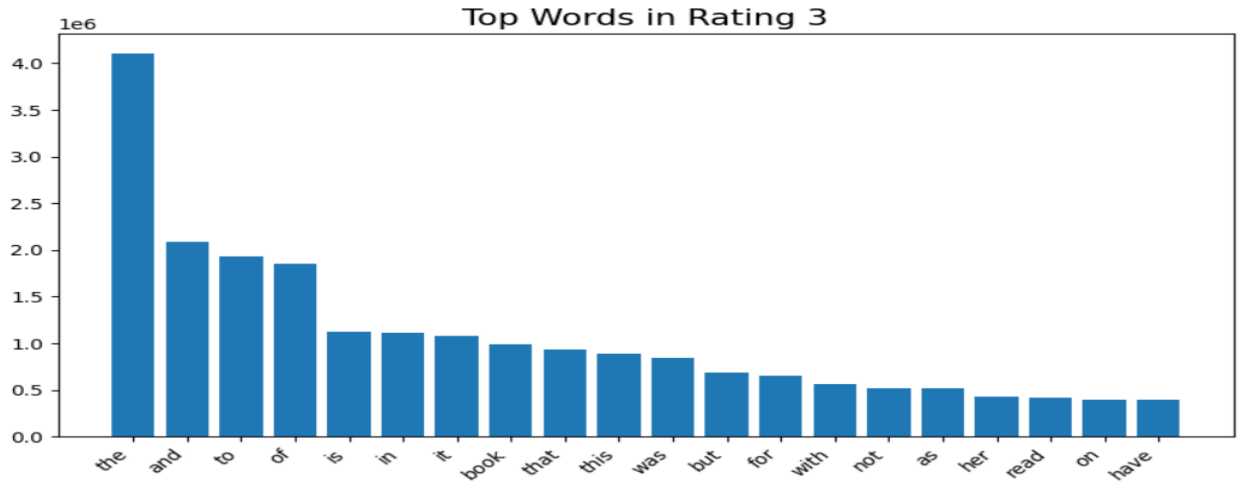
- **Top Common and Unique Words:**



[illegible]

Word cloud visualization of the text data. The words are arranged in a dense, overlapping manner, with colors ranging from green to purple. The most prominent words are 'bennounne', 'chalea', 'strings', 'bradbury', and 'yillustr'.





7. Dataset Creation

The initial dataset exhibited imbalanced data, meaning the distribution of classes (star ratings) was not equal, with some classes having significantly more samples than others. This is a common

issue in machine learning as models tend to perform poorly on minority classes, leading to biased predictions.

To address this, a balanced dataset was prepared with the following target distribution for ratings:

- **1-star: 350000**
- **2-star: 470000**
- **3-star: 620000**
- **4-star: 800000**
- **5-star: 980000**

8. Samples

The documentation includes sample reviews for each star rating (1-5) to illustrate the nature of the data after preprocessing. These samples provide concrete examples of the textual content and their corresponding ratings, aiding in qualitative understanding of the dataset.

Showing 10 sample review(s) for Rating 1:

1. what a bore
2. i had to really struggle to finish this the plot is lame the characters unbelievable and the dialogue actually the whole story flow is very stilted worst of all is the fact that this work is part of a series and it shows characters relationships situations etc pop up helterskelter some authors of series books go overboard in explaining things from the previous books in the series but this author did the extreme opposite she gave absolutely no background
3. goodkinds books are uneven some really good some not so good this work is hands down his worst a real stinker do not buy it under any circumstances dont even check it out of the library as it is a waste of your valuable time
4. couldnt get beyond the ridiculous scenario of remy finding a chained lion during a zombie attack that becomes her pet kitty the amount of suspension of disbelief for me as a reader was impossible the author has no concept of making connections so that her readership is pulled into her fictional world this book is for the immature reader caught up with action for its rush alone the writing is atrocious and characterization horrific
5. if you are a fast reader with patience try it i purchased it to read with my book club and they eventually chose not to read it i couldnt bring myself to read this thick book so i gave it away at a charity event

6. disgust book ever read tell disciplin kid put right track kill spirit make robot follow parent best interest child jesu send parent hell best exampl use love violenc teach someon someth never laid hand rod anyon even save prostitut stone next time parent want beat child sin rememb sinner appli rod first good parent good children

7. first grisham read partner love proceed plow everi one book last five six year save brethren book becom noth essay go nowher read broker last juror thinnest plot call plot confid say would buy anoth grisham book store publish world burn someon give one away may think im harsh get feel grisham heart longer write publish hip pocket

8. found book entertain easi read shotti histori david barton use primari sourc fabric histori want told sourc use splice manipul way show african american make deal devil democrat portray republican jesu assum democrat love slaveri republican nota person research histori would advis reader book accept everyth print fact look exampl david barton use preston brook dsc beat charl sumner illustr evil democrat parti southern democrat quick violenc northern republican due slaveri know charl sumner first took seat unit state senat democrat massachusett stori continuesin bleed kansa crisi border ruffian approach lawrenc kansa sumner denounc kansasnebraska act crime kansa speech may may two day sack lawrenc sumner attack author act stephen dougla illinoi andrew butler south carolina compar butler quixot dougla sancho panzasumn said dougla present chamber noisom squat nameless anim proper model american senat also portray butler taken mistress though ugly other alway love though pollut sight world chast sighti mean harlot slaveri sumner threehour orat later becam particularli person insult mock yearold butler manner speech physic manner impair stroke butler suffer earliersen stephen dougla illinoi also subject critic speech suggest colleagu sumner orat damn fool sumner go get shot damn foolpreston brook nephew senat butler infuri sumner attack uncl first brook intend challeng sumner duel end brook consult fellow south carolina repres laurenc keitt duel etiquett keitt instruct duel gentlemen equal social stand suggest sumner occupi lower social statu compar drunkard due supposedli coars languag use speech brook conclud turn sinc sumner social stand would appropri beat canetwo day later afternoon may brook confront sumner sat write desk almost empti senat chamber brook accompani chamber keitt henri edmundson virginia brook address sumner mr sumner read speech twice care libel south carolina mr butler rel mine sumner began stand brook began beat sumner sever head thick guttapercha cane gold head could reach feet sumner knock trap heavi desk bolt floor brook continu bash sumner brook rip desk floor time sumner blind blood stagger aisl collaps laps unconsci brook continu beat motionless sumner cane broke point left chambernon stori use david barton book set record straight american histori black white tell one portion stori lead reader one side stori without tell realli lead alterc use illustr say accept one person doctrin fact david barton done use multipl primari sourc discov histor fact avail use credibl secundari sourc discov truth

9. great book and story i like all the characters and the ending was very exciting i want to read more books about these characters

10. i didnt really find this book very stimulating

Showing 10 sample review(s) for Rating 2:

1. celtic blood is an independent novel by james john loftus it tells the story of morgund macaeth in thirteenth century scotland the book begins with seaward a viking being washed ashore and the only one to survive the wreck of the ship he was on he is adopted by the macaeth clan where he grows into a fearsome warrior politics in those times are much more direct and a band of men attack the leader of the people who took him in and kill him seaward escapes and then is faced with the challenge of protecting morgund who is the macaeth heir i had a challenging time at the beginning of the book trying to decide if james style was brilliance or just a refusal to follow the bounds of the english language while there are times that description spins into something that is more poetry than prose the sad truth is that i found myself wanting to hear more about the parts that he skims over and less about the things he describes in detail the time and setting are given their due but even well into the book i didnt know the characters as people i also found the panegyric description got in the way of me just following the plot i love beautiful description as much as the next reader but i also love knowing what is happening that the description is attached to between the sentence fragments and dizzyingly quick point of view shifts i couldnt follow the story as much as i wanted to love this book i cant recommend it

2. im a big fan of alternative historyharry turtledoves guns of the south got me interested in the genre id read sm stirling before conquistador the peshawar lancers and really enjoyed him so when i started his nantucket series i was expecting a good read which they are and arent the premise of the nantucket series is that the island of nantucket is inexplicably hurtled back in time to the bronze age the islanders must figure out how to survive and interact with this strange new world dies the fire is a companion novel to the nantucket series you neednt have read the trilogy to understand whats going onit just lets you in on a few characters mentioned in the other books dies starts the night of the event when nantucket disappears tho these characters dont know that and suddenly anything remotely electrical stops working batteries die cars wont run even gunpowder wont explode any more the hero mike havel is a bush pilot flying a rich family to their place in idaho when their plane just quits midair he manages to bring the plane down in one piece but the mother is injured pretty badly after discovering that nothing works the party sets off in search of helpcivilization theyve got two things going for themmike is a combat veteran and knows how to survive in the woods and the youngest daughter astrid is a fantasyloving tolkien freak who has her own extremely wellmade bow and arrows and knows how to use them meanwhile in corvallis oregon juniper mackenzie a folksingerwiccan priestess is performing in a tavern when there is a blinding light and then all is dark except for the fires flaming out of control from a that crashed in the middle of town juniper her deaf daughter eilir and their friend dennis realize something very wrong has happened and head for the hills literally the rest of the book is how the two groups grow in size try to avoid plague cannibals and mad warlords and eventually come together a pretty good tale of survival but while the plot is sound the whole book felt strained one of an authors goals should be for the reader to connect with his or her characters and i just couldnt i cared very little for what happened to mike juniper or any of the numerous supporting cast i think the only one i

really felt anything for was Astrid and that's mainly because I'm a Tolkien freak too. Also, I understand that in a postapocalyptic world such as this, life is going to be mean, nasty, brutish, and short, but I don't need explicit descriptions of this every other chapter. Sometimes every chapter, most of the people who die, and trust me, a lot of people die, do so in extremely horrific ways which the author seems to spend entirely too much time describing to the reader between the cannibals and sadistic biker bicycles, not motorcycles, gangs, there's a lot of raping, blood, and body parts, and chalk it up to me being a new mother, but I got awfully tired of hearing about children being killed or dying in other ways, maybe once, OK, too often, and I started just skipping whole sections of chapters. I don't read horror novels for a reason. Finally, there's the whole Wiccan storyline which, after a while, started to sound more like proselytizing than part of the story. All the good guys are either agnosticatheistic or pagan, the Christians are all bigots or lapsed, even the sole Buddhist ends up joining Junipers Coven. As for the epilogue, that just got a little too out there, causing me to ask myself, is this book about an alternative history or sword-and-sorcery fantasy? In the end, I'm not sure I can recommend this book. It left a rather sour taste in my mouth and mind. The Peshawar Lancers, sure, at least I don't remember it being this gruesome, even Conquistador, but let the fire die out on the bookshelf.

3. spoiler alert: this book is written as a love song about the author's dear friend who died of the rare form of cancer. The main character succumbs to it in the story, but Green takes what is at best a short story or novelette and fills it with chapters of back story and characters that are not necessary to the book. Do we really need to know about Walters' marriage to an Asian mailorder bride? Do we need the details about Ed's exwife and son? All of the extra fluff distracts from the devastating loss. The book is about, I could not summon up the emotion I should have been feeling for the characters in this book.

4. child abuse is a horrific reality in our world, and it needs to be exposed. Shiho Kishimoto makes an attempt to do this, but does more to mussle the waters than to awaken us. It's a pity because the subject needs far better and believable treatment. The translation undoubtedly is part of the problem, but the author doesn't seem to have a firm grasp on the gravity of the subject. The heroine seems self-serving and just is not likeable. The book does have a powerful start, but gets lost, and just is not the book to stir the reader. Definitely not recommended. It's a shame.

5. the sex was good, but not earth-shattering. This book started out great and got boring in the middle. By the end, I was mostly skimming the pages just to see how it ends. It was anticlimactic. This is the first book by this author that I have read. I hope they get better.

6. started out loving the book, but actually skimmed the last pages just to see if anything unpredictable happened. I don't usually review a book that I don't like, but with the cost of books and the lack of free time to read, I really wished it had been better. I read to be transported to another place, and this book did not do it.

7. the book is very tiny. This is for small children, so they need big books to see things better. Very disappointing.

8. begin note love lion alrassan quit fond fionavar enjoy good amount sail sarantium felt dirti read part detest lord emperor made feel need bathein word thrill ggk abil breath life charact build emot realiti gratuit violenc sex disgust repetit phrase repetit phrase basic wildli disappoint interpret ggk intend emot tug tigana go way could feel sorri brandin dont care know fionavar cute refer though poor silli tragic beauti dianora torn credit card bill paper shredderso let see ive got straight daddi send child go neighbor kingdom beat submiss child goe oop kingdom fight back oop win hack wrist children mani peopl kingdom hideous tortur princ defend peopl year wipe whole popul read brandin incap take respons send child battlefield begin withand dianora find endlessli sexi clever loveabl suppos feel bad men loyal alberico may twice evil question doesnt justifi brandin blecha devin vaunt memori whose memori mean much memori import repeat devin memorywho vaunt memori fail figur dianora dianora play memori come use remark disappointmentand gratuit sex incest violenc mind tast romanc novel read overload seem titil nauseat readerggk write fabul well convey tortur emot well wrote touchingli sweet final scene repeat incred well mani time devin reflect burden alessan like count number time saddest tale long tale told appear fionavar triologyah well captur essenc exil well wasnt worth slog get itin opinion

9. i understand the overall high rating as the story is very good and well written however not captivating as it is spoilt with trivia of a tokyo tourist guide and an obsession with judo and karate techniques real pity

10. i struggled to finish this book the premise was farfetched and the characters acted stupidly there was one slight twist to the story but even that was fairly predictable i wont read any others in the series

Showing 10 sample review(s) for Rating 3:

1. this frustrating but ultimately pretty decent sequel to the fascinating city of ember picks up right after the finish of the first volume as the entire populace of the dying underground city make their way out of harms way into the upper world sparks covers the travails of the next few months as the emberites encounter a small town struggling to get by with a population even smaller than the survivors of the underground world there are two major problems with the book i think first that it is far less original in feel and setting the postapocalyptic worlds of films like the mad max series and countless sf novels from the past century arent ever very far off and second the whole premise of the arguments between the two communities rests on the notion that the leaders of sparks are arrogant and unwilling to speak with the emberites as equals or even really as other intelligent people while it is true that the emberites are ignorant of many things such as seasons which become a big plot point that is not dealt with particularly realistically its also clear that they have their own special knowledge and yet they act demoralized and the people of sparks act superior from the getgo the apocalypse is only two centuries past surely someone besides the two young heroes of the first book lina and doon would have some notion of cooperation still the book has some narrative drive ms dupraus pacing is fairly solid and she manages to keep up interest in the plot

despite the basic storyline originality issues i have with it and she also succeeds i think in creating some fairly interesting characters maddy mrs hester without seeming to try very hard though i didnt like this nearly as much as the first book it still showed plenty of promise which alas the last two books didnt really live up to for me still ms duprau has some interesting ideas and is a fairly compelling writer and im curious to see what she does next

2. there were parts of this book i truly liked it caught my interest at the very beginning and drew me in i read until the very end i liked some of the characters a lot the only problem is that later in the book i was hating some of these same characters or not giving a flying leap about them because they werent developed fully the cover art was fabulous i liked the gorillas and they made too brief an appearance in the book i liked that the book was relatively short i liked that i didnt pay very much for it there were things that made me want to throw my kindle against the nearest wall the technology presented in supervirus came across as very implausible to me i am not a computer genius or a techno geek but other authors have presented technical jargon to me in novels and made it believable also the military actions in the book seemed ad libbed and explanations too abbreviated really one of the worst endings i can remember with all of the above written most of which does not sound positive i gave this book stars i understand that mitchell has written nonfiction before concerning environmental issues and has been involved with primate groups the sections of the book concerning these subjects are much more plausible and enjoyable i am assuming because of this knowledge i also understand that he is a physicist and researcher though and that makes me wonder about my other issues with the book i think andrew mitchell has a lot of potential writing fiction and i will try his next book one of the huge problems with ebooks is that they are put before the reading public open for mass criticism before some of the background editing and revision that used to be done on books prior to publication is accomplished so mitchell and the reading public gets to read all these comments which would have come from copy editors or technical editors in a prior world new authors get published sooner but the books stories are usually not as polished as in the past goodbad i guess we as the reading public get to decide i do believe we get to read works of raw talent that might have never made it past the front desks at a lot of publishing companies previously sometimes i am in the mood for that and sometimes i am not

3. the jury is still out its very well written but i am still not sure if the characters are more human than insect or vice versa a bit hard to create true imagery

4. wasnt one of my best efforts think want something between books i am sure i will recommend read hickam coalwood book meantime october sky coalwood way better lot of world building

5. overlong musing looking at himself ad nauseum sorry expected more from a pulitzer prize for fiction however good character development

6. well enough written story should have been more interesting lots of reparation but goes quickly

7. not one of rankins best the usual grungy witty conversations are delicious but after half way they are taken over by a borstal pederasty storyline which has been played out recently in a tv episode cant remember whether it was lewis sherlock cumberbatch jonathon creek or even morse
8. set in early san francisco this is an interesting tale of mystery and murder it is the first book in the series
9. well written mint baldacci
10. the title of this book is very deceptive although the book does give you much interesting information about hindu history and culture it only lightly touches on the million gods only gods to be exact brahman and ganesha all chapters are an average of three pages long and there are a few chapters that are fairly assigning such as the chapter titled the manly englishmen and the effeminate hindu it was more about the author saying that indians are manly based on their reproduction rate and englishmen arent because they only use guns and their birth rate is declining waste of a chapter poor argument and who cares over all an ok read at best but you will not learn about the million gods no pictures of them no details and brief chapters

Showing 10 sample review(s) for Rating 4:

1. i cant believe how much i enjoyed this story it was both nostalgic in setting but also in languagea contemporary plot design added a fresh pageturning effect that did indeed have me grabbing every bit of reading time i could to finish my only questionwhats with the five time use of preternatural sort of felt like a word of the day effort but it didnt bother me a bit a very enjoyable read
2. frankli book great space odyssey fact wonder anybook match still love book fiction read sever time book almost everyth one ask ps may read never read think arthur c clark blackmail write book
3. this collection begins with an interesting history about the authors and making of the curious george series geared more for the adult reader than the child it is followed by five separate curious george stories i downloaded the curious george collection for my daughter who has a whole collection of curious george paperback books she is in kindergarten and is able to read the stories but they tend to run long so she tires and needs me to help her finish the kindle version sports the familiar illustrations although they are in black and white sometimes an illustration in the paperback version would span two pages which makes it awkward in the kindle version because my daughter only sees half the illustration on a page at a time she understands what is happening because she has the paperback version at home but this can be a problem for young children otherwise this is a great collection for the price and a wonderful one to have when traveling with children
4. needed it for school

5. loved the story ms steele s ability to make the reader feel drawn into the story leaves you looking forward to her next book
6. love the way the author intertwined the present with the past so subtly u almost dont see it com it coming great job great read
7. this is a story of a boy whos parents die and he goes to live with his aunt but runs away he runs into a bunch of people who he makes a difference in their life also he does things that people in the history of the world would not be able to do it gets so that people who dont even know him personally know his name in the end he does something that nobody would even think about if you would like to know you will just have to read it for yourself
8. wonder witti thought provok look manner time two world war set mostli franc studi american expatri revolv around elliot social connector view nonfrench societi barbar larri wwi fighter pilot look death face decid want live differ peopl meet great question aris stori public imag possess happi planet book start slow pick charact start interact long section end author also charact stori retel convers larri larri hindu belief section expositori bog content fairli interest
9. china in the st century provides a good background on china though it is rather bland reading in addition to providing background on china wasserstrom also addresses important issues eg is war likely over taiwan in a evenhanded manner the author also brings appropriate focus eg reminding readers that even though intensively competitive about of its largest businesses are still stateowned however the book lacks any solid projections for the future especially chinas economics
10. i enjoy reading the inspector lynley series in this story the inspectors life take a dramatic turn its quite sad

Showing 10 sample review(s) for Rating 5:

1. this was a fascinating book it was a quick read as i did not want to put it down it raised interesting questions regarding the nature vs nurture debate
2. i like this book it was entertaining and i liked the quirky characters if you enjoyed the outlander characters this is worth a read
3. amazing things about cooking you didnt knowreally
4. bukowski is a writer you either get or you dont get most people claim hes far too harsh and that his prose is too vulgar well women isnt a book for the squemish its basically a follow up to post office its long after the protagonist henry chinaski bukowski has established himself as a poet and living the life of some moderate fame and success he still drinks excessively and messes around with mad women the most impressionable being lydia who is insane hes always attracting the mad ones this book was very enjoyable though i think only fans of bukowski will appreciate its verse it helps if you are a misanthrope

5. i love these books they are one of my favorite series i hope they make some more movies to go along with the books

6. as with all of danielle steeles books once i start i have to finish this story had you wondering all the way through just how it would turn out in the end just lovely

7. this continues the excellent series with gabriel allon proving once again that he is a most human hero i enjoy the way the author brings the action to a high level without losing the depth of feelings the hero and his team experience a wonderful read

8. warning this is not some sugarcoated fairy tale its brutally honest and very emotional its a tellitlikeitis story about loyalty and selfsacrafice depressing not at all just intensethe story is told through the eyes of a boy called fitz who has very little time to be a boy at all at six he is taken from his mother who we know very little about to live with his fathers stableman burrich as he grows his grandfather king shrewd decides to have him educated by the time he turns he knows how to write read maps fight and of course assassinate sound like a lot for a kid to deal with it is this is a story about his struggles and experiances in his service to the king i recommend it to anyone whos tired of stupid harry potter knocoffs and ready for something different

9. the fourth book in the kathleen turner series doesnt disappoint it was so good that i read the entire book over christmas break in one day i have to say until this book i was unsure which team i was pulling for but after this book i am definitely team kade the action and suspense keeps you turning the page and needing more i love this series and i have already preordered the fifth book and purchased blanes story a definite mustread series love it

10. i have to thank my friend cheryl cooper for this book i love it when my friends send me recommendations this book was a fun read even though it is part of a series i never was lost the characters are brilliant and the story was a great mix of humor and mystery i am looking forward to reading more from this author

9. Train-Test Split

In machine learning, the dataset was split into:

- Training Set: Used to train the model (e.g., 80% of data).
- Testing Set: Used to evaluate the model's performance on unseen data (e.g., 20% of data). This split helps ensure the model can generalise effectively to new, unseen reviews. The 80-20 split is a commonly used ratio that balances the amount of data available for training with a sufficient portion for independent evaluation.

#stratified train test split

```
X_train,X_test,y_train,y_test=train_test_split(X,y_categorical,test_size=0.2,stratify=y_categorical,random_state=42)
```

10. Bidirectional LSTM (BiLSTM)

The Bidirectional LSTM (BiLSTM) model was selected for this task due to its effectiveness in processing sequential data like textual reviews.

- **Captures context in both directions:** Unlike standard LSTMs, BiLSTM reads the text sequence both forward and backward. This is crucial for understanding sentiment in reviews, where the meaning of words can often depend on their preceding and succeeding context.
- **Effective for sequential data:** Customer reviews are textual sequences. LSTMs are inherently well-suited for handling long-term dependencies within such sequences, significantly outperforming simpler Recurrent Neural Networks (RNNs).
- **Proven performance:** BiLSTM models have demonstrated higher accuracy in sentiment analysis and review rating prediction tasks compared to traditional machine learning algorithms or simpler neural network architectures.

11. Model Training Logic (including BiLSTM-specific Preprocessing)

Before feeding the data into the BiLSTM model, the following additional preprocessing steps were applied, building upon the general data cleaning:

11.1. BiLSTM-Specific Preprocessing Steps:

1. **Text Cleaning:** All text was converted to lowercase, and special characters and punctuation were removed. This ensures a clean input for tokenisation.
2. **Truncating Excessively Long Reviews:** Reviews exceeding 300 words were truncated to reduce noise from overly verbose inputs and ensure a uniform input size for the model.
3. **Encoding Labels:** The star ratings (1–5) were first label-encoded and then converted into one-hot vectors. This is a standard practice for multi-class classification problems when using neural networks with a softmax output layer.
4. **Tokenization and Padding:** Text was tokenized using Keras' Tokenizer, converting words into numerical sequences. These sequences were then padded to a fixed length of `max_len=200`. Padding ensures that all input sequences to the neural network have the same length, which is required for batch processing.

5. **Optional Stopword Removal & Lemmatization:** These steps were considered and applied to further reduce vocabulary size and potentially improve semantic understanding by reducing words to their base form.

11.2. BiLSTM Model Architecture and Training: The BiLSTM model was trained using the following architecture and regimen:

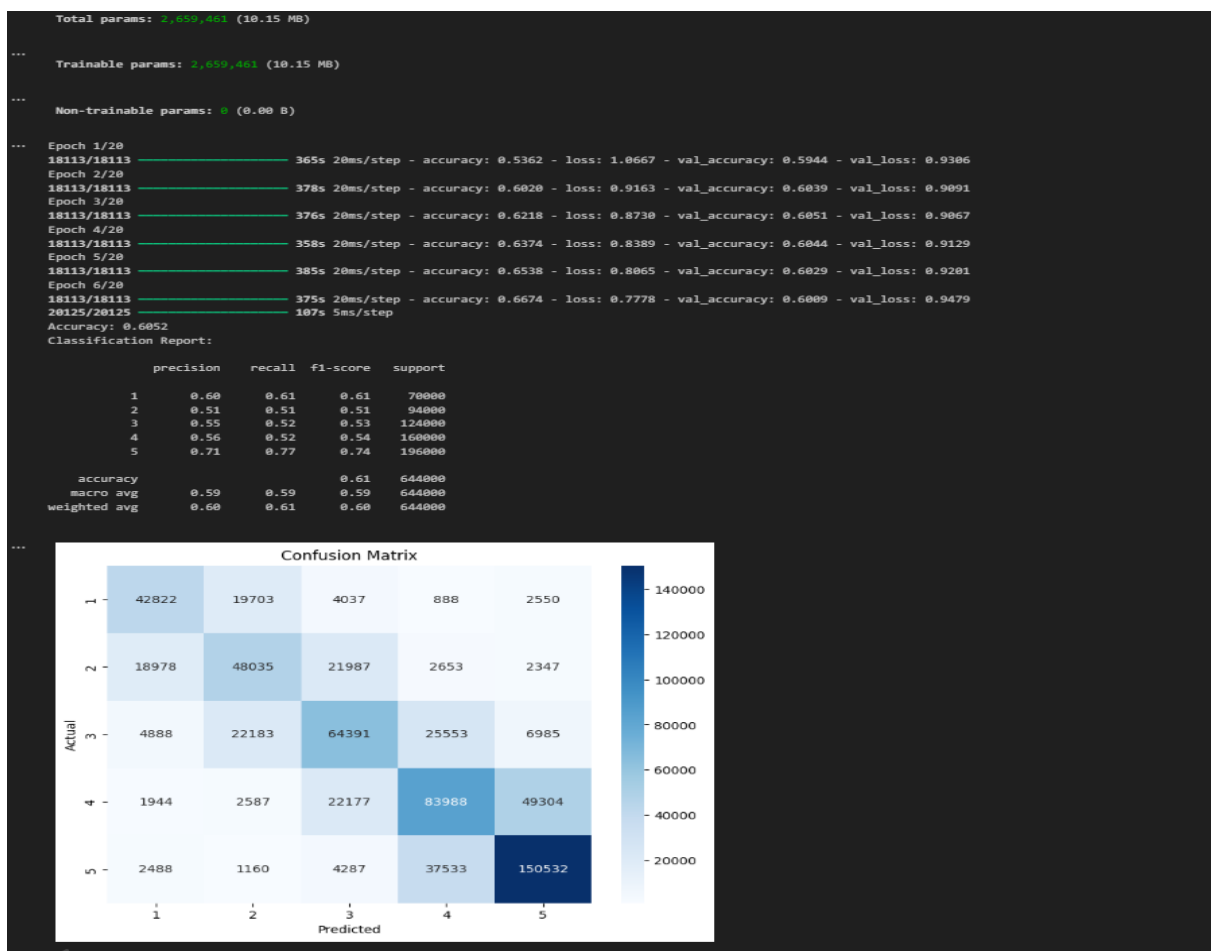
- **Embedding Layer:** Words were converted into dense vector representations. This could be achieved using either randomly initialised embeddings or pretrained embeddings (e.g., GloVe). The choice of pretrained embeddings typically helps the model leverage pre-existing linguistic knowledge.
- **Bidirectional LSTM Layer:** This core layer consisted of 128 units and incorporated dropout and recurrent dropout. The bidirectional nature allows the network to process the sequence from both forward and backward directions, capturing more comprehensive context. Dropout mechanisms help prevent overfitting.
- **Dropout Layer:** A 50% dropout rate was applied after the BiLSTM layer. This technique randomly sets a fraction of input units to 0 at each update during training, which further aids in preventing the model from becoming overly reliant on specific features and improving generalisation.
- **Dense Output Layer:** The final layer comprised 5 neurons with a softmax activation function. This configuration is standard for multi-class classification, where each neuron corresponds to one of the five star ratings, and softmax ensures the output probabilities sum to 1.
- **Loss Function:** categorical_crossentropy was chosen as the loss function, which is suitable for multi-class classification where labels are one-hot encoded.
- **Optimizer:** The Adam optimizer was used for efficient convergence during training. Adam is an adaptive learning rate optimisation algorithm that has proven effective across a wide range of deep learning problems.
- **Callbacks:** EarlyStopping and ReduceLROnPlateau were implemented to dynamically adjust the learning rate and prevent overfitting during training.
 - EarlyStopping monitors a validation metric (e.g., val_loss) and stops training if the metric stops improving for a specified number of epochs (patience).
 - ReduceLROnPlateau reduces the learning rate when a metric has stopped improving, allowing the model to make finer adjustments.

Training was performed using a stratified 80-20 train-test split to ensure that all rating classes were proportionally represented across the training and testing datasets.

12. Evaluation Results and Interpretation

Upon completion of training, the model's performance was evaluated on the unseen test set. The evaluation involved several key metrics:

- **Accuracy:** This metric measures the overall proportion of correctly predicted ratings.
- **Precision, Recall, F1-Score:** These metrics were evaluated per class to assess how effectively the model predicted each individual rating (1-5 stars). This is crucial for imbalanced datasets, as high overall accuracy might mask poor performance on minority classes.
- **Confusion Matrix:** A confusion matrix was utilised to visualise misclassifications between different ratings, providing granular insights into specific errors made by the model. It highlights which classes are most often confused with others (e.g., 1-star reviews misclassified as 2-star).



13. Additional Considerations

13.1. Future Work:

- **Explore advanced models:** Investigate Transformer-based models (e.g., BERT, RoBERTa) which often achieve state-of-the-art performance in NLP tasks.
- **Ensemble methods:** Combine multiple models to potentially improve predictive accuracy and robustness.
- **Expand data sources:** Incorporate reviews from a wider variety of domains and languages to enhance generalisation.
- **Hyperparameter optimisation:** Conduct more rigorous hyperparameter tuning to find optimal model configurations.
- **Explainability:** Implement techniques to understand *why* the model makes certain predictions, improving trust and interpretability.

13.2. Conclusion:

This project successfully developed an AI-based system for automated review rating using a Bidirectional LSTM architecture. Through meticulous data collection, cleaning, and specific preprocessing steps, the system aims to provide an efficient and standardised method for sentiment analysis from textual reviews. While the core methodology is robust, further refinement and detailed performance reporting will enhance its practical application and scientific contribution.